



Key considerations of commercial vehicles

- **Gross Vehicle Weight (GVW):** GVW determines the vehicle's capacity for carrying goods or passengers.
- **Fuel Type:** Commercial vehicles can be powered by diesel, petrol, CNG, or electric.
- **Payload Capacity:** This is the maximum weight a commercial vehicle can carry, for simpler understanding, when the vehicle is used for transporting goods.
- **Wheel base, Number of wheels/tires (known to use radial or bias-ply construction)**
- **Ground clearance:** the distance between the ground and the lowest point of the vehicle, measured in millimeters.
- **Steering type:** Hydraulic power assisted
- **Safety systems:** ABS, ESC, Designed for driver comfort, Build quality to mitigate crash impact
- **Key specifications** include engine type, capacity, power output, torque, fuel tank capacity, transmission, brakes, suspension, tire size, wheelbase, and overall dimensions (length, width, height).
- **Awareness and Responsiveness to** ensure safe, well-maintained and optimally performing vehicle, assist and safety systems

Sensitized and Effective decision making

Driving condition responsiveness refers to how quickly and appropriately a driver reacts to changing road and traffic conditions. This includes their reaction time to hazards, their ability to maintain control of the vehicle, and their decision-making processes in varying scenarios. Factors like stress, fatigue, and driver ability can significantly impact this responsiveness.

Key Aspects of Responsiveness

Enforcement Directive networked
Tachograph for Driving Time, Working Time
and Rest periods

Reaction Time
Control over the vehicle
Defensive Driving practices
Sensitized or Effective Decision-making

Key Factors affecting Responsiveness

Driver Training/Evaluations
Driver Ability/Anticipatory Guidance
Driver Stress
Driver Alertness / Fatigue
Road and Traffic conditions
NSSR-RS Training Programme, Pre & Post ride
checklists, instructions
Enforcement Directives for DT, WT and Rest
DT, WT and Rest Timings surveyed per state
and across state borders

Defensive Driving (Areas of importance for safer and sustainable commuting by commercial vehicles)

- ☐ Safe and Fuel-efficient driving
- ☐ Awareness, knowledge and understanding of technical characteristics, benefits and operations of safety controls
- ☐ Limits and conditions for the use of brakes
- ☐ Better use of speed and gears
- ☐ Ways of slowing down and braking on downhill stretches, and factored decisions or actions to be taken in the event of a mechanical or active safety system failure
- ☐ Ability to help passengers find possible ways for safety and effectively use public & private infrastructure
- ☐ Dedicated Lane driving
- ☐ Managing conflicts between safe driving and other roles as a driver
- ☐ Interactions with passengers
- ☐ Knowing and understanding specific needs of certain passengers/ and target groups like children, the disabled, the commuters needing alpha assistance
- ☐ Knowledge and maximum working periods (Driving time, working time and rest) with the use of a tachograph

Defensive Driving (Areas of importance for safer and sustainable commuting by commercial vehicles)

- ☐ Knowledge of passenger expectations, luggage, need for safety equipment, need for seat belts, lap belts and other anchorage
- ☐ Knowledge of vehicle load rules and causation factors
- ☐ Awareness of traffic signs, road traffic risks, types of accidents in the transport sector and driving condition responsiveness, accident statistics, accident causation factors, consequences to vehicle, functional expectations, equipment, organizational and public human resources, materials, money involved
- ☐ Ability to respond to emergencies, concerning situations, accident-causing factors via training, and responsiveness for procedures of First Aid, Fire Evacuation/ Fire Detection & Response potential driver / passenger conflict situations, implications to the safety of passengers/commuters and the impactful issues for road safety
- ☐ Understanding of the importance of physical and mental wellbeing for road safety,, with the opting for “of a healthy lifestyle/diet, and avoidance of alcohol, drugs and other substances, and interests for decisions” known to affect behaviour/alertness, the adherence to work-rest cycles known to cause fatigued conditions

ii. **Active Safety Systems that universally help**

- ☐ Electronic Stability Control (ESC)
- ☐ Tyre Pressure Monitoring System
- ☐ Cruise Control / Adaptive Cruise Control
- ☐ Advanced Braking System / ABS
- ☐ Collision Avoidance Warning Systems / Forward Collision Warning Systems
- ☐ Lane Departure Warning Systems
- ☐ Driver Alert Control Systems
- ☐ Reverse Assist Systems
- ☐ Fuel Monitoring System

☐ Charter for Driver Fitness/Driver Training, Vehicle Fitness and Performance Analysis, Road System understanding, Alpha Assistance

☐ Periodic Inspection

☐ Pre-ride and Pot-ride checklists for fitness and system analysis

☐ Corrective Maintenance

Responsiveness for A-S-P and DI in operational environment
A: Awareness
S: Sensitization
P: Preparedness
DI: Deep Interaction

Types of buses

- City Buses
- Inter-city Buses
- Suburban Buses
- Tour Buses
- Staff & Contract Buses
- School Buses

Key considerations for buses

- **Primary Function:** Passenger transport.
- **Capacity:** High passenger capacity, often with multiple rows of seats.
- **Features:** Comfort-focused seating, often with amenities like air conditioning and entertainment systems.
- **Value addition:** Fuel Monitoring System, Cruise Control, Gear Shift Advisor, Firmware over the air, Telemetry
- **Safety features:** Blind Spot Mirror, Anti-skid flooring, Panic button, Guard Rails, Emergency brakes with ABS

Value addition functions in buses

- ☐ Fuel Monitoring System
- ☐ Improved Engine Hood Installation
- ☐ Pagoda Gear Box
- ☐ Gear Shift Advisor
- ☐ E-Viscous Fan for engine cooling
- ☐ Cruise Control
- ☐ Gateway Domain Controller Unit for seamless communication, data exchange, and security
- ☐ Firmware over the air
- ☐ Fleet Management Telemetry
- ☐ USB Charging Ports

Need for Intelligent Transport Systems, Navigation, Perspective Vision and Feedback Systems that improve road system understanding for concerns like height and weight restrictions for vehicles, planning for pretexts such as bridges, tunnels, ring roads, flyovers, under-passes, need for emergency call features on route

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Safety system functions in buses

- ☐ Top Marker Lamp
- ☐ Blind Spot Mirror, Indirect Vision Detection
- ☐ Heavy Duty Wiper
- ☐ Emergency Exit, Service Doors, Peep Windows
- ☐ Transparent Door
- ☐ Guard Rails
- ☐ Anti-skid flooring
- ☐ Panic button
- ☐ Safe Seat Upholstery with Grip Handle
- ☐ Seat Belt Anchorage, Lap Belt, Impact Shield
- Instead of Air bags
- ☐ Hand hold Straps
- ☐ LED Destination Board

Safety system functions in buses

- [] Fog Lamp
- [] Reverse Parking Sensor
- [] 7 Layer Exhaust Insulation
- [] Fire Detection and Alarm System Indicators (in Engine Compartment, Passenger Areas and Luggage Areas)
- [] Fire Detection Systems with Cable test for satisfactory behaviour, Repel test for insulation, Burning material behaviour tests
- [] All Wheel Disc Brakes
- [] Emergency Braking with ABS
- [] Gear Level with Cable Shaft
- [] Speed Deceleration Stop Lights
- [] DRL Head/Tail Lamps
- [] Unitized wheel bearing
- [] Robust Multi-piece Propeller Shafts